

across all ten tasks, matching DFORL and outperforming both ∂ ILP and NeuralLP. These results underscore ANDRE’s effectiveness in symbolic environments and confirm its capacity to induce interpretable logical rules from examples. Moreover, the subrules extracted by ANDRE for each task consistently adhered to the two syntactic constraints, demonstrating the model’s built-in capability for learning rule structures.

4.2 LARGE-SCALE KNOWLEDGE BASES

In this section, we test ANDRE on three large-scale knowledge bases, including Countries dataset (Bouchard et al., 2015), Unified Medical Language System (UMLS), and Nations dataset (Kok & Domingos, 2007). Following Gao et al. (2022), the Countries dataset is divided into three different subcategories ordered by learning difficulty: S1, S2, and S3 sub-datasets. Statistical details of the dataset are deferred to Appendix I. Following the propositionalization step, a trainable dataset is extracted from each knowledge base, based on the target head predicate. Then, each dataset is split into a train and a validation set. After training, three different performance metrics are computed: Accuracy, Mean Reciprocal Ranks (MRR), and HITS@K, where $K \in \{1, 3, 10\}$.

Table 2 compares the performance of ANDRE with baselines, including NTP λ (Kok & Domingos, 2007), NeuralLP, and DFORL. Based on all metrics, ANDRE outperforms the baselines in all datasets, except Nations - *blo* and UMLS - *intw* datasets. In general, the performance of ANDRE is either comparable or better than DFORL as the strongest baseline, confirming the reliability and scalability of ANDRE. The extracted rules for each dataset and predicate are presented in Appendix I. Moreover, we computed and compared the training and evaluation time for ANDRE and the baselines in Table 7 (Appendix J), which shows that ANDRE is significantly faster than DFORL, though it is not the fastest method. Based on Appendix K, ANDRE also outperformed the baselines on the UW-CSE (Davis et al., 2005) dataset and has shown competitive performance on the Alzheimer-amine dataset compared to the strongest baseline.

4.3 SYNTHETIC PROBABILISTIC DATA

As ANDRE is specifically designed to handle probabilistic (non-Boolean) data, classical ILP benchmarks—with their crisp symbolic representations—are not ideal for fully evaluating its capabilities. To address this, we construct synthetic datasets tailored to assess ANDRE’s ability to learn interpretable logical programs under uncertainty. Unlike traditional ILP benchmarks, these datasets allow controlled manipulation of data properties such as noise, ambiguity, and rule complexity, thereby providing a more rigorous and scalable testbed for neuro-symbolic methods like ANDRE.

We generated a diverse collection of synthetic datasets, each corresponding to a logical rule of varying complexity and number of subrules. All data points were sampled from a uniform distribution and split into training and validation subsets using the same underlying distribution to ensure consistency. A fixed set of random seeds was used throughout all experiments for reproducibility. Each dataset consists of the probabilistic valuations of m candidate body predicates with the corresponding Boolean label derived from a target logical rule (see Appendix G for an example of how h is determined).

We generated three different datasets, based on the following hierarchical rules with increasing complexity: $R_1 \leftarrow b_1 \wedge \neg b_9$, $R_2 \leftarrow R_1 \vee (\neg b_2 \wedge b_8)$, and $R_3 \leftarrow R_2 \vee (b_3 \wedge \neg b_5 \wedge b_7)$. In each case,

Table 2: Performance comparison across knowledge bases. ACC@ h means the accuracy of the model on a dataset with head predicate h . Predicates *blo*, *int*, *neg*, and *intw* stand for *blockpositionindex*, *intergovorgs3*, *negativecomm*, and *interacts_with*.

Dataset	Metrics	NTP λ	NeuralLP	DFORL	ANDRE
Countries	ACC@S1	100.0	100.0	100.0	100.0
	ACC@S2	100.0	100.0	100.0	100.0
	ACC@S3	100.0	–	100.0	100.0
Nations	MRR	0.418	0.565	0.789	0.831
	HITS@1	41.79	52.49	73.88	79.52
	HITS@3	41.79	60.95	84.58	84.73
	HITS@10	41.79	61.19	85.07	93.81
	ACC@blo	100.00	50.00	100.00	99.06
	ACC@int	84.62	84.62	84.62	89.38
	ACC@neg	37.50	75.00	75.00	89.24
UMLS	MRR	0.301	0.667	0.750	0.772
	HITS@1	29.95	61.27	71.41	69.28
	HITS@3	30.11	72.31	78.82	81.19
	HITS@10	30.11	72.31	78.97	83.36
	ACC@isa	65.96	63.83	91.48	94.53
	ACC@intw	83.67	86.67	100.0	98.97

the model must infer the correct rules from probabilistic input features only with semantic losses. We only benchmark ANDRE against DFORL, as the only publicly available and most competitive method on classical ILP datasets and knowledge bases.

Based on Table 3, ANDRE consistently outperformed DFORL by 21.63% on average. According to the extended Table 9 in Appendix L, ANDRE successfully extracts the target logical program when provided with a sufficient number of labeled examples. From the perspective of rule complexity, ANDRE remains robust: as complexity increases from R_1 to R_3 , the drop in accuracy for DFORL is $5.92\times$ greater than that observed for ANDRE, highlighting the latter’s resilience to rule complexity.

Noisy Synthetic Data. To assess ANDRE’s robustness in the presence of noisy labels, we conducted a series of experiments using probabilistic datasets with intentionally corrupted labels. In these datasets, a controlled percentage of both positive and negative labels were randomly flipped, simulating real-world scenarios with mislabeled data. While the complexity of the target rules was kept consistent across experiments, the number of training samples was large and fixed for each rule to eliminate underfitting as a confounding factor.

Table 3: Validation accuracy on complex synthetic datasets with varying number of subrules (compressed version of Table 9 and 10).

Dataset	Sample Size Range	Noise Range	Average Accuracy	
			DFORL	ANDRE
R_1	20 – 200	—	0.85	0.92
R_2	50 – 1000	—	0.73	0.94
R_3	200 – 2000	—	0.70	0.90
R_4	200	10 – 30	0.61	0.71
R_5	500	5 – 45	0.52	0.71
R_6	1000	5 – 25	0.48	0.75

The target rules used in this study are structurally similar to those in the previous experiments but drawn from a separate hierarchy: $R_4 \leftarrow \neg b_1 \wedge b_9$, $R_5 \leftarrow R_4 \vee (b_2 \wedge \neg b_8)$, $R_6 \leftarrow R_5 \vee (\neg b_3 \wedge b_5 \wedge \neg b_7)$. Table 10 (Appendix L) reports the performance of ANDRE and DFORL across varying levels of label noise, while Table 3 compresses the results. On the synthetic probabilistic benchmarks (R_1 – R_6), ANDRE consistently achieves higher predictive accuracy and substantially improved rule extraction quality compared to DFORL. In particular, while DFORL collapses under moderate label noise due to the use of product-based fuzzy operators, ANDRE remains stable, correctly recovering 9 out of 14 ground-truth rules across noisy settings. Although performance degrades for the deepest rule (R_6) at the highest noise level, ANDRE still extracts significantly more correct symbolic structure than competing differentiable ILP methods. These results empirically confirm that attention-based operators preserve dominant predicates under uncertainty, enabling more reliable rule induction from probabilistic data.

Ablation Study. To assess the impact of the attention-based logical network on ANDRE’s performance, we conducted an ablation study by comparing the standard ANDRE model to a variant in which the attention-based conjunction and disjunction operators were replaced with product-based alternatives. The experimental setup mirrors the evaluations in Table 9 (Appendix L), and we report the average test accuracy for both configurations. As shown in Table 4, the attention-based operators improve accuracy by 21.05%, demonstrating their critical role in effective rule learning.

Overall, the strong performance of ANDRE arises from its carefully designed architecture. The introduced rule space allows flexible inclusion or exclusion of body predicates, even in the presence of ambiguity. Unlike product-based operators, ANDRE’s attention-based conjunction and disjunction mechanisms maintain full differentiability, preserve accuracy, and alleviate the vanishing gradient problem common in fuzzy logic reasoning. Additionally, the syntactic loss components encourage syntactically valid rule structures, even when semantic signals are noisy or degraded.

Table 4: Ablation Study Results.

Method	ANDRE w/o Attention	ANDRE
Accuracy	0.76	0.92

5 CONCLUSION

We introduced an *Attention-based Neuro-symbolic Differentiable Rule Extractor (ANDRE)*, a novel neuro-symbolic ILP method that formulates rule learning as optimization over a continuous rule space. By leveraging soft attention mechanisms for conjunction and disjunction, ANDRE enables differentiable learning of symbolic rules without requiring predefined rule templates. The model identifies the most relevant predicates through a tailored learning strategy and enforces logical con-

sistency using specific syntactic loss functions. Experimental results across classical ILP benchmarks, knowledge bases, and synthetic probabilistic datasets demonstrated ANDRE’s strong performance in both rule extraction and classification accuracy. Notably, ANDRE remains robust in the presence of noise, successfully extracting symbolic rules under up to 25% mislabeled data depending on the task complexity. Compared to state-of-the-art data-driven ILP methods, ANDRE consistently outperforms in terms of accuracy, interpretability, and stability. These findings highlight ANDRE’s potential as a reliable and interpretable component in neuro-symbolic learning pipelines.

Limitations and Future Research. Despite its strong performance, ANDRE has several limitations. First, while the proposed continuous rule space enables flexible predicate selection, the current architecture is primarily designed for relatively shallow rule structures and may struggle to efficiently represent more complex or deeply nested logical compositions. Second, the model relies on propositionalization of relational data, which may limit its ability to fully exploit rich relational structure in large knowledge graphs. Finally, symbolic rule extraction depends on confidence thresholds over learned distributions, which may introduce sensitivity in borderline cases. Future work will focus on extending the framework to support more expressive rule compositions and developing more principled mechanisms for direct relational reasoning without full propositionalization.

REFERENCES

- Kamal Acharya, Iman Sharifi, Mehul Lad, Liang Sun, and Houbing Song. Integrating neurosymbolic ai in advanced air mobility: a comprehensive survey. In *Proceedings of the Thirty-Fourth International Joint Conference on Artificial Intelligence*, pp. 10362–10370, 2025.
- Samy Badreddine, Artur S. d’Avila Garcez, Luciano Serafini, and Mike Spranger. Logic Tensor Networks. *Artificial Intelligence*, 303:103649, 2020. URL <https://api.semanticscholar.org/CorpusID:229678739>.
- Guillaume Bouchard, Sameer Singh, and Theo Trouillon. On Approximate Reasoning Capabilities of Low-Rank Vector Spaces. In *AAAI spring symposia*, 2015.
- Brandon Curtis Colelough and William Regli. Neuro-Symbolic AI in 2024: A Systematic Review. *ArXiv*, abs/2501.05435, 2025. URL <https://api.semanticscholar.org/CorpusID:274180938>.
- Andrew Cropper and Sebastijan Dumančić. Inductive logic programming at 30: a new introduction. *Journal of Artificial Intelligence Research*, 74:765–850, 2022.
- Andrew Cropper and Rolf Morel. Learning programs by learning from failures. *Machine Learning*, 110(4):801–856, 2021.
- Andrew Cropper and Stephen H Muggleton. Logical minimisation of meta-rules within meta-interpretive learning. In *Inductive Logic Programming: 24th International Conference, ILP 2014, Nancy, France, September 14-16, 2014, Revised Selected Papers*, pp. 62–75. Springer, 2015.
- Jesse Davis, Elizabeth Burnside, Inês de Castro Dutra, David Page, and Vítor Santos Costa. An integrated approach to learning bayesian networks of rules. In João Gama, Rui Camacho, Pavel B. Brazdil, Alípio Mário Jorge, and Luís Torgo (eds.), *Machine Learning: ECML 2005*, pp. 84–95, Berlin, Heidelberg, 2005. Springer Berlin Heidelberg. ISBN 978-3-540-31692-3.
- Luc De Raedt and Kristian Kersting. Probabilistic inductive logic programming. In *Probabilistic inductive logic programming: theory and applications*, pp. 1–27. Springer, 2008.
- Honghua Dong, Jiayuan Mao, Tian Lin, Chong Wang, Lihong Li, and Denny Zhou. Neural Logic Machines. In *International Conference on Learning Representations*, 2019. URL <https://openreview.net/forum?id=BlxY-hRctX>.
- Richard Evans and Edward Grefenstette. Learning explanatory rules from noisy data. *Journal of Artificial Intelligence Research*, 61:1–64, 2018.